Ex0: Python Toolchain Setup

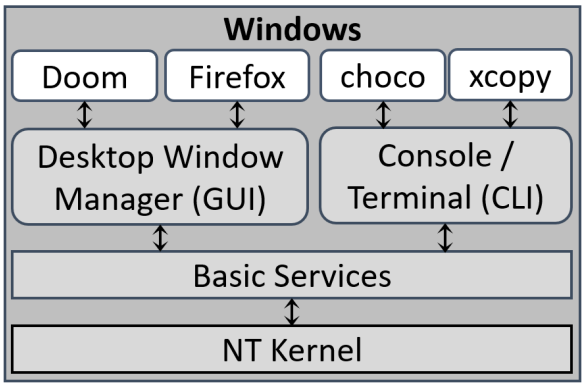
Overview

This exercise is intended to help students set up your toolchain so that you can build projects the course. This will involve setting up the runtime environments and IDE for Python 3.10. You are not required to use the IDE we specify, though it is recommended. There are no deliverables and there is no grade for this exercise.

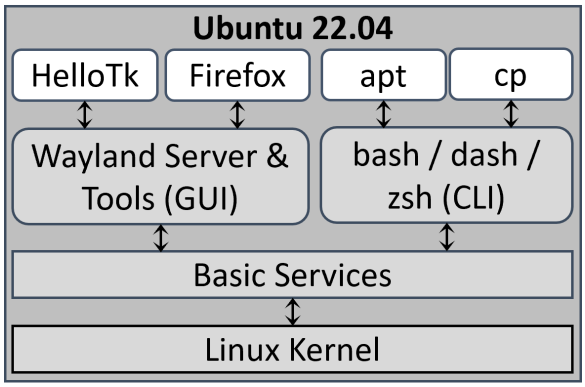
The environment specified in this document will be the one exclusively used for testing student projects. Note that these instructions are written for Windows users, but students using Ubuntu 22.04 and other Linux distributions should be able to set up their environments for the course. With that said, please note that there is no instructional support for other operating systems. Macintosh computer users should consider using a virtual machine running Ubuntu 22.04 or installing BootCamp and Windows 11 (which is free to UF students). If you are using Windows 10 all updates must be installed; you may need to re-install Ubuntu within WSL.

Architecture

Windows and Ubuntu are both built as end-user operating systems. Typically, a modern user-facing operating system (OS) includes a command line interface (CLI) and graphical user interface (GUI):

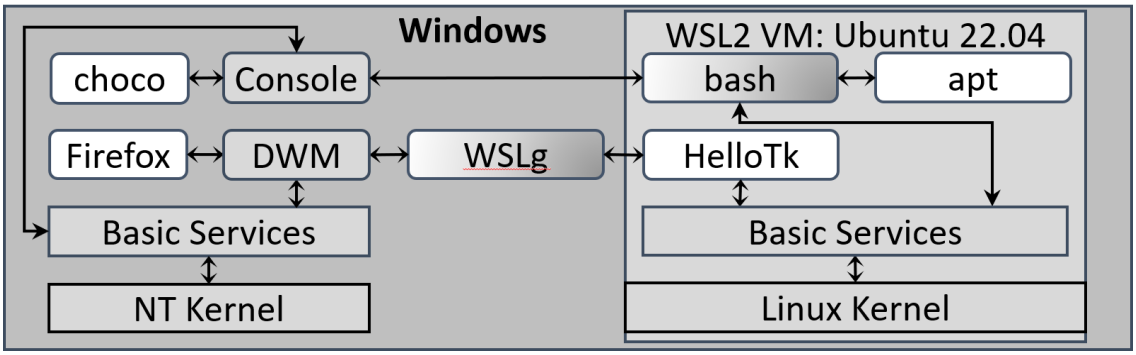


Windows 10 Runtime Architecture



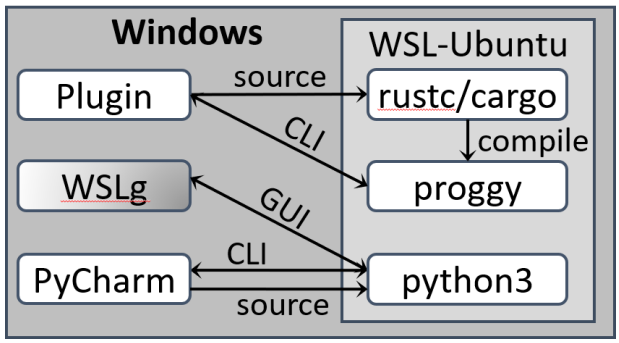
Ubuntu 20.04 Runtime Architecture

Windows can run Linux-based operating systems in the Windows Subsystem for Linux (WSL):



WSL2 and interactions between kernels and major system components.

Windows has built-in support for graphical Linux applications. This is useful because Python's runtime and development tools are most well developed on Linux systems. Additionally, it provides us a common target for testing. Many IDE suites can also integrate with WSL:



GUI software & WSL2 on Windows

In this exercise, students will set up Ubuntu 22.04 on WSL2. They will also setup the PyCharm IDE suite and the Rust plugin to communicate with WSL. Projects and exercises in this course will target this environment.

Tasks

This exercise is broken into these steps:

a) Install Chocolatey package manager and other tools
b) Setup WSL2 with Ubuntu 22.04 and Python 3.10
c) Setup PyCharm and Rust Plugin (or your own IDE)

Before starting this process, make sure your Windows installation is up to date – some commands require the latest patches / updates to function properly! This is particularly true of WSL, which is in active development.

WSL & Chocolatey Package Manager

To install enable WSL and install Chocolatey, open PowerShell (🔎 → "**`PowerShell`**") as an Administrator (right-click → "Run as Administrator"). From there, run these commands (copy-paste):

```
wsl --install
```

```
Set-ExecutionPolicy Bypass -Scope Process -Force; [System.Net.ServicePointManager]::SecurityProtocol = `
[System.Net.ServicePointManager]::SecurityProtocol -bor 3072; iex ((New-Object `
System.Net.WebClient).DownloadString('https://chocolatey.org/install.ps1'))
```

You can test this with the command "**`choco -?`**". Once it is ready, install the other tools:

```
choco install pycharm
```

If you prefer to use IntelliJ + Python Plugin, or no IDE, you may skip these packages – but this is what we will test on!

WSL Ubuntu Setup

Open Ubuntu **from the start menu** to select a username / complete installation. Then update and install tools:

```
sudo apt update
sudo apt upgrade
sudo apt install build-essential cmake gdb python3-pip python3-virtualenvwrapper python3-tk \
                 python3-venv rustc x11-apps
```

Once this is done, add a few settings to allow WSL, the IDEs, and other tools to connect seamlessly:

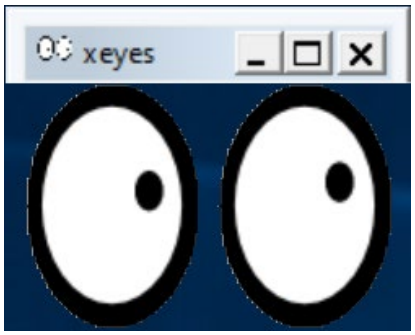
```
echo -e "export VIRTUALENVWRAPPER_PYTHON=$(which python3)" >> ~/.profile                #1
echo -e "source /usr/share/virtualenvwrapper/virtualenvwrapper.sh" >> ~/.profile         #2

wget https://raw.githubusercontent.com/JetBrains/clion-wsl/master/ubuntu_setup_env.sh && \
bash ubuntu_setup_env.sh && rm -f ubuntu_setup_env.sh                                    #3
```

These lines do the following:

1) Set the python version for virtual environments (VIRTUALENVWRAPPER_PYTHON)
2) Enable virtual environment tools for Python (virtualenvwrapper.sh)
3) Set up the SSH connection for CLion and C/C++ remote execution / debugging

GUI Testing

Close all terminals (consoles) and then run **`xeyes`** to test the server connection. You should see this:

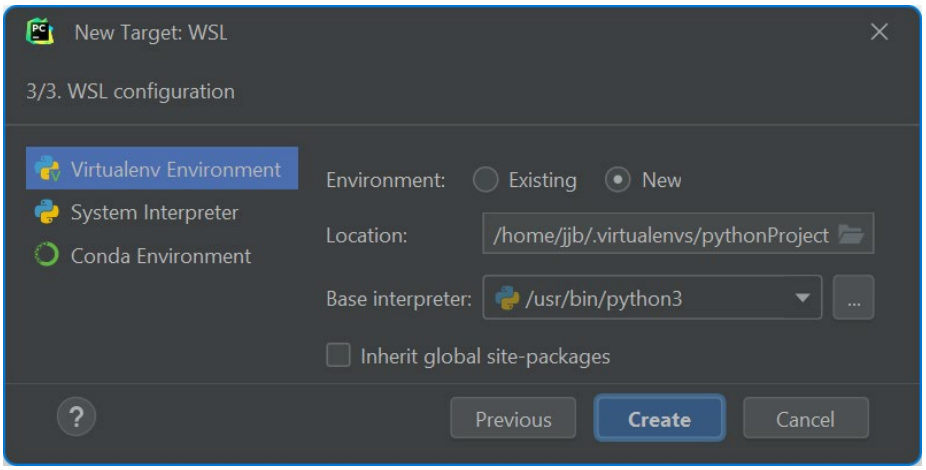


IDE Setup

This section covers the setup of PyCharm IDE and the Rust plugin.

Setup PyCharm IDE

1) Open **File | Settings | Project | Python Interpreter | Add Interpreter | On WSL**.
2) Once Ubuntu is found, click **Next**.
3) The Python path should be **`/usr/bin/python3`**. Click **Create**, then **OK**.
4) Load & execute the "HelloTk" project to test the setup.



Setup Rust Plugin

1) In PyCharm, open **File | Settings | Plugins**. Type "Rust" in the search box; the plugin should appear.
2) Click "Install". Afterwards, you will need to agree to restart by clicking "Restart IDE".
3) After the IDE restarts, open **File | Settings | Languages & Frameworks | Rust**.
4) The plugin should have detected your Rust installation in WSL (e.g., **\\wsl$\Ubuntu-22.04\usr\bin**).